Mach Number 2.75, Altitude 6300m, AoA 2 degrees

Volume Mesh:

Cell Quality [Measure : Orthogonal]

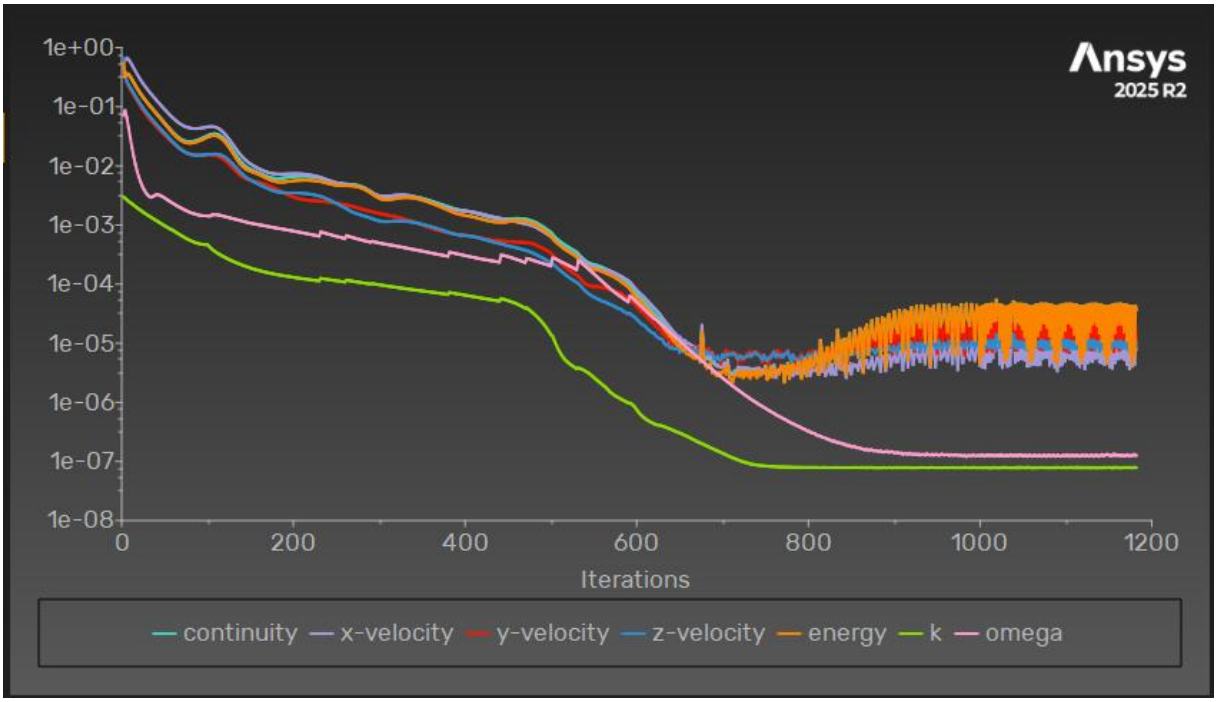
Minimum = 0.031273331, Maximum = 0.99999236, Average = 0.91496978.

Marking Cells in Range (0.031273331 0.081273331)...

Info: Total marked entities in the range is 367, is limited to 49 by max number of allowable cell clusters (10).

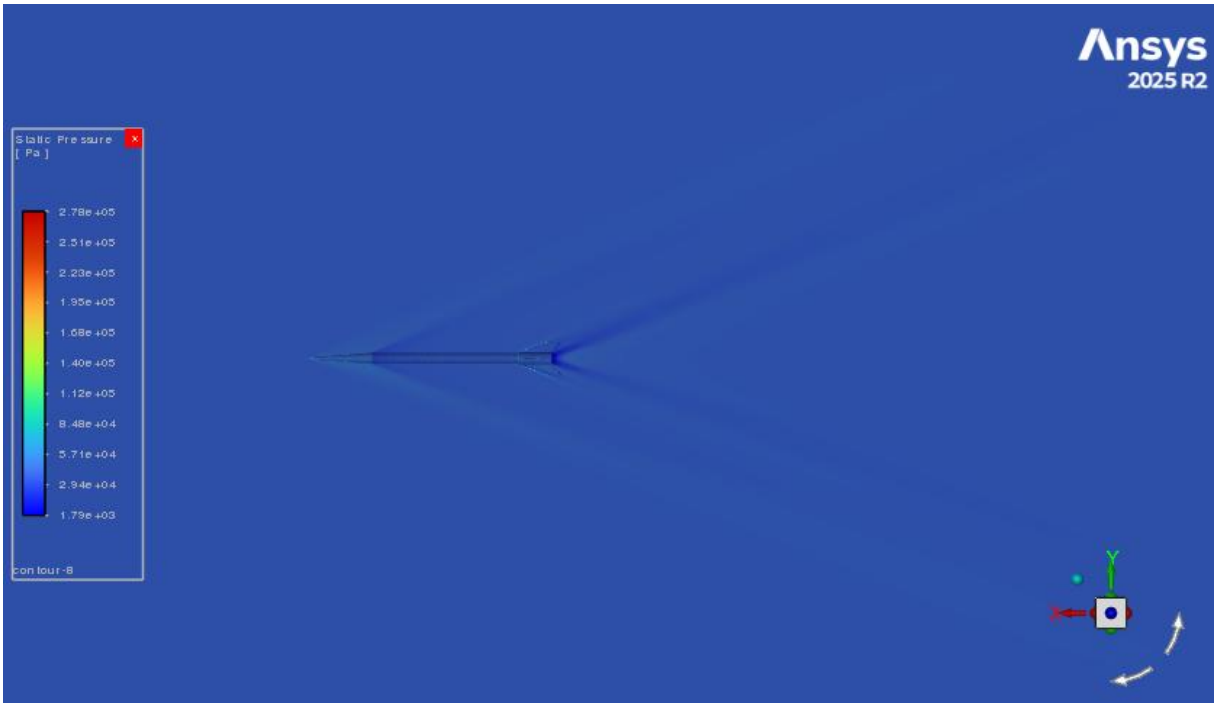
Results:

Residuals:

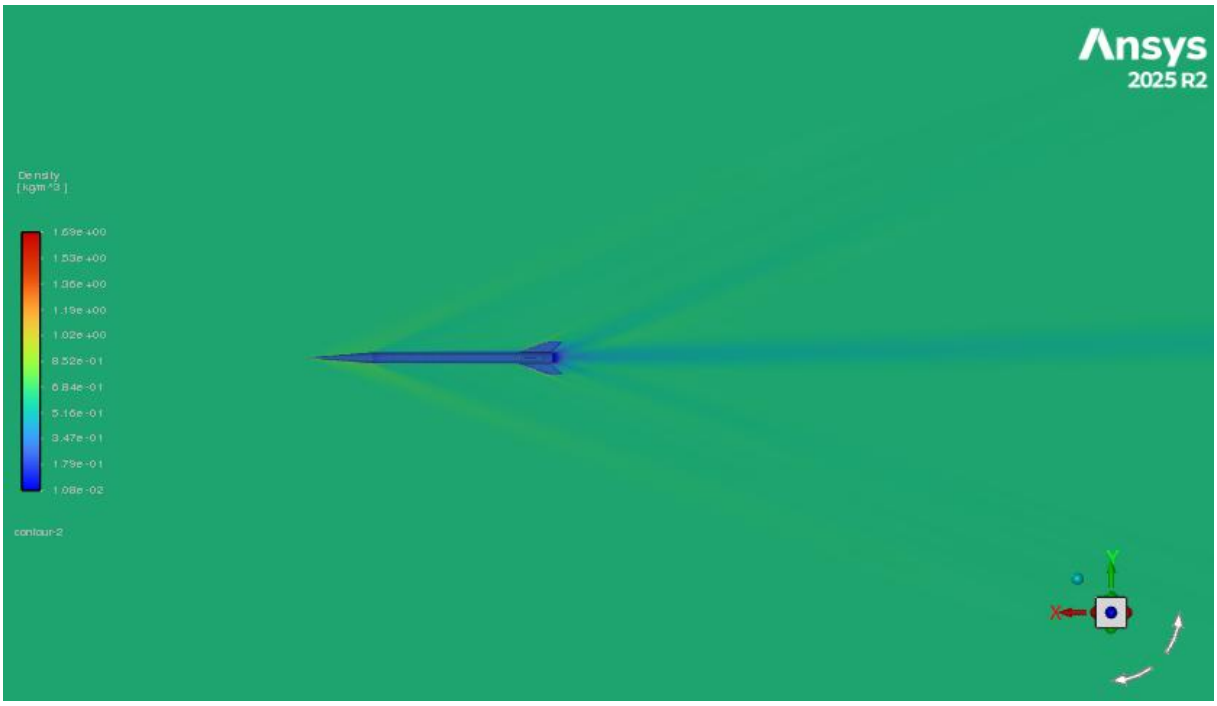


Loose convergence is likely due to the y+ - value > 100 in combination with the k-w SST and the 367 low orthogonality cells. Further refinement was not possible as the mesh (7.5 million cells) was at the limit of my computational power.

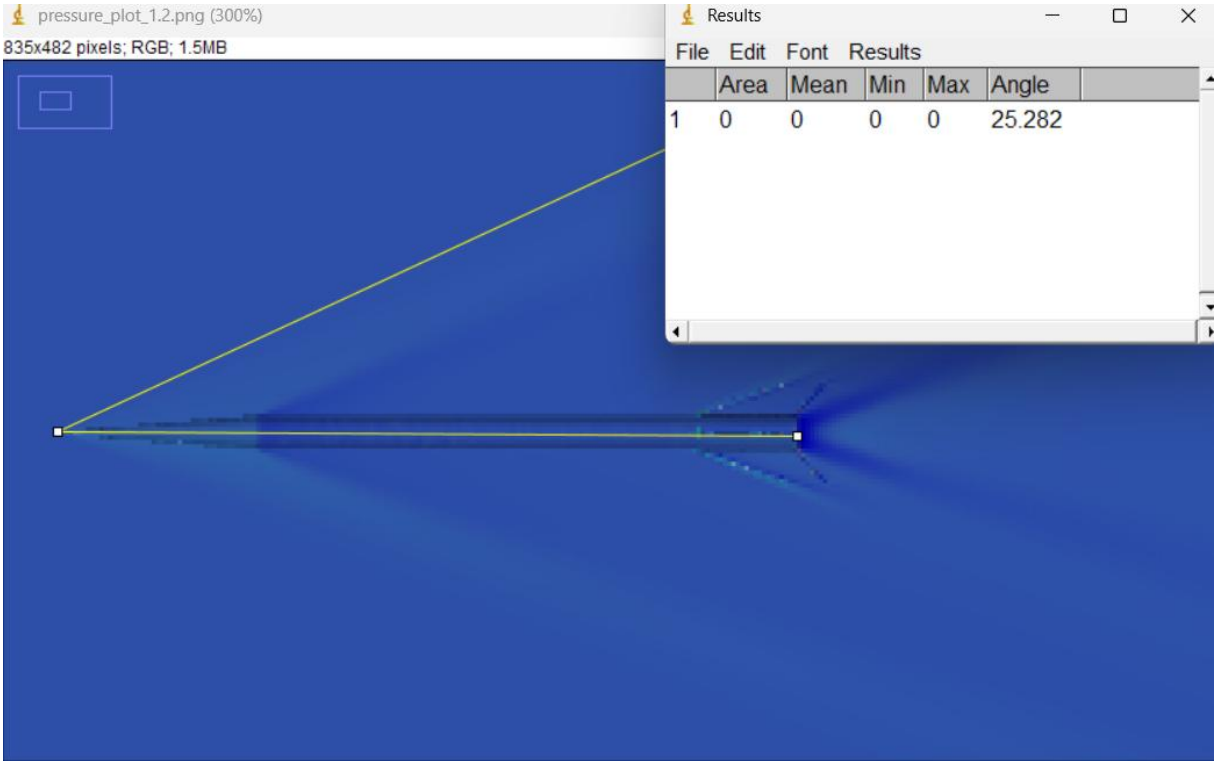
Pressure Plot:



Density Plot:



Validation:



Calculation:

Nosecone angle ~ 4.7° degree, angle to flow ~ 2.7° (AoA = 2°)

$$\tan\theta = \frac{2\cot\beta(\mathrm{Ma}_1^2\sin^2\beta - 1)}{\mathrm{Ma}_1^2(\kappa + \cos 2\beta) + 2}$$

- ⇨ Shock angle ~ 23.1° (referenced to flow)
- ⇨ Shock angle with reference to x-Axis ~ 25.1°

Error ~ 1.8% (Keep in mind that the shock angle is not perfectly measured)